Lab 1 - Final Task

Extracted from the final mini-project/task section at the end of the notebook.

Source notebook: Lab1.ipynb **Extracted cell:** 60

9. Final mini-project

Complete the final mini-project:

1. Load the Iris dataset into a DataFrame.
2. Display the first 10 rows, shape, columns, missing values, and descriptive statistics.
3. Create at least two new derived columns.
4. Create three filtered DataFrames.
5. Use `groupby()` to compare at least two features by species.
6. Create one Matplotlib scatter plot and one Matplotlib histogram.
7. Create one Seaborn boxplot and one Seaborn heatmap.
8. Create or improve an OOP class with at least four methods.
9. Write two short paragraphs: one about the most important data pattern and one about how this lab prepares you for future AI topics.